

Minimum Cost Analysis of Feeder Routing in Distribution System Planning

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Abstract: A new approach for optimal network routing in distribution planning is presented in this paper. The distribution network routing planning problem is formulated as a non-linear constrained, non-differentiable optimization problem. The main objective is to minimize the total cost which is the summation of reliability costs, feeder resistive loss, investment and maintenance costs. Simulated annealing, which is a general purpose combinatorial optimization algorithm, is used to obtain a global optimum solution of the problem.

I. INTRODUCTION

Value based planning in distribution systems has recently been receiving more attention due to increased investment costs and the necessity to quantify and justify the reliability levels in a system. Utilities faced with increasingly limited resources, strive to maintain high levels of reliability by adopting improved methodologies in planning, operation, construction, and maintenance [1]. An important question in this activity is what is the required level of reliability. Many utilities now recognize that the total system cost used in the decision making process must include the value to the customers in the form of power interruption costs, in addition to investment, maintenance and resistive loss costs.

From an economic perspective, any reliability criterion, and the resulting design, should be based on the cost of providing extra reliability by changing one or more of the system parameters versus the benefits accruing to society from the additional reliability. The basic concept of reliability cost/worth evaluation [2,3] is portrayed graphically in Figure 1. There is considerable literature on the subject of optimum distribution planning [4-15]. Some of the important objectives of optimum distribution planning are

- (i) Providing maximum possible reliability and reduced interruption costs
- (ii) Minimizing the investment and maintenance costs
- (iii) Minimizing power loss and achieving acceptable voltage levels.

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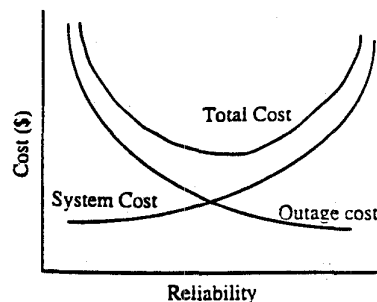


Figure 1. Variation of Total Cost With Reliability

The factors of reliability, system costs, voltage quality and losses are directly related to the network configuration. The financial justification of the solution in each sub problem is related and therefore these factors should be considered in an integral evaluation.

Various mathematical programming techniques, such as integer programming, mixed integer programming, transportation, transshipment and branch and bound methods have been applied to the problem of feeder configuration optimization [7-15]. Many of the existing methods consider the optimum routings minimizing only the system costs which include investment costs of distribution feeder sections, distribution equipment, maintenance costs and feeder resistive loss costs. While there has been growing interest in value based power system planning, there exists relatively few publications which consider reliability costs in total distribution cost optimization. This paper presents an approach to the minimization of total costs, i.e., summation of outage, resistive loss, investment and maintenance costs in order to obtain optimum routes for primary and lateral section distribution feeders. The problem is modeled as a non-linear, non-differential and constrained optimization problem. A general purpose combinatorial optimization algorithm, known as, simulated annealing is used to solve the problem. Simulated annealing has been applied successfully to network routing problems in the area of CAD [16-18]. The main advantage of this algorithm is the ability to include many realistic constraints and non-linear objectives. It can be used to solve a mixed integer non-linear programming problem to obtain a global optimal solution.

Performing a minimum cost analysis requires an assessment of the costs of providing reliable service and quantification of the worth of having it. The annual expected values of the customer cost of interruption and curtailments

can be added to the predicted annual capital, feeder resistive loss and maintenance costs to form a total cost index for the specified project. Comparisons of alternatives are then made on the basis of the total cost. In this way unreliability is considered as a part of total cost. An important task therefore is to determine the worth or value of system reliability. There have been considerable efforts to quantify the worth of service reliability [19-27]. A variety of approaches have been used to determine the actual or perceived costs of customer interruptions. One method which has been used [20-27] to establish acceptable reliability worth estimates is to survey electrical consumers in order to determine the monetary losses associated with supply interruptions. Interruption costs data obtained from a customer survey can be used to develop customer damage functions. A customer damage function is an estimate of the customer interruption cost expressed as a function of interruption duration. A contingency enumeration approach is given in [28] to calculate outage costs in distribution systems. The influence of operating practices on the outage costs is illustrated in [29]. The general application of value based planning is illustrated in [19-21]. This paper considers the distribution feeder routing problem in static expansion planning, i.e., with a single planning period, and can be extended to dynamic expansion planning [5,13].

II. Problem Formulation

The considered objective is to obtain the optimum routings of distribution feeders and the location of substations which minimize outage costs, cost of feeder resistive loss, investment and maintenance costs. Thus the objective function is both non-linear and non-differentiable. This can be expressed as follows.

$$\min \text{ cost} = \sum_{k=1}^{nk} P_k \left\{ \sum_{j=1}^{ne} \sum_{i=1}^{np} L_{ik} \cdot C_i(r_j) \cdot \lambda_j + K_c \cdot \sum_{l=1}^{nf} r_l \cdot \frac{(P_l^2 + Q_l^2)}{|V_l|^2} \right\} \\ + \Sigma \text{ Investment Cost} + \Sigma \text{ Maintenance Cost}$$

subject to the following constraints

$$P_l^{\min} \leq P_l \leq P_l^{\max}$$

$$Q_l^{\min} \leq Q_l \leq Q_l^{\max}$$

$$V_l^{\min} \leq |V_l| \leq V_l^{\max}$$

$$I_l^{\min} \leq |I_l| \leq I_l^{\max}$$

Where,

nk - number of the load steps

ne - number of contingencies

np - number of load points

P_k - probability of load step k

L_{ik} - load at load point i for step k

r_j - interruption duration of contingency j

C_i(r_j) - cost of interruption j for the duration r_j at load point i for the specified type of customer at i

λ_j - failure rate of distribution element (primary, lateral feeder section, transformer etc.)

K_c - resistive loss coefficient

nf - number of feeder sections

V_l - voltage at the beginning of feeder section l

P_l, Q_l - active and reactive powers in feeder section l

r_l - resistance of feeder section l

III. Solution Algorithm

Simulated annealing is a general optimization approach based on randomization techniques for solving combinatorial optimization problems. This algorithm has been applied successfully in many areas [16-18, 31]. It is based on the analogy between the simulation of annealing of solids and the problem of solving large combinatorial optimization problems. The total cost formulated in the previous section is referred to as the energy function. A parameter called "temperature" is defined in this technique which is of the same dimension as the energy. The system to be optimized starts at a high temperature and is cooled down, until the system freezes and reaches the global optimum. This is similar to the annealing of a crystal during growth to reach a perfect structure. At each temperature, a new random configuration is generated by perturbing the current configuration. It has been shown that this algorithm can provide a global optimum. The details of the algorithm can be found in [16-18]. The main advantage with this algorithm is that the objective function can be complex, non-linear and non-differentiable which is true in the case of the distribution network routing problem. A brief description of the algorithm used is given as follows.

- 1) Input the, system data, reliability data, interruption cost data, initial temperature for simulated annealing, random number seed, cooling rate, the step size of the control setting and the number of moves at each step.
- 2) Design a global cooling schedule $T_{i+1} = \alpha(T_i) T_i$, where $\alpha(T_i)$ is the cooling rate. At each temperature T_i , a set of moves n_i are performed. For each move do the following steps
- 3) Select a random configuration, g_{i+1} connecting the load points. Conduct a load flow for all the load steps for this configuration and if the constraints are satisfied perform the following steps or select another random configuration. The load flow technique described in [32] was used as it is very efficient for radial distribution networks and it provides a numerically robust solution. The required equations can be written in the forward or

- backward directions, to compute voltage, reactive power and active power on the different branches.
- 4) Compute the feeder resistive loss and the outage costs for the given configuration for the given load level. Compute the energy function (E) which is the total cost formulated in the previous section. The change in energy function is computed. as., $\Delta E(g_i) = E(g_{i+1}) - E(g_i)$.
 - 5) Update the system configuration
if $\Delta E(g_i) \leq 0$
or
 $e^{-\Delta E/T_i} > \text{random}[0,1]$
where $\text{random}[0,1]$, is a random number uniformly distributed in the interval [0,1].
 - 6) Use the cooling schedule with each temperature T_i and perform a number of moves n_i . A new configuration for each move is considered and a load flow using forward and backward updates is obtained. If any constraint is violated, ignore the configuration or use the next configuration and go to Step 4
 - 7) If the stop criterion is not satisfied, then the system is not yet frozen and the process is continued by returning to Step 2.
The initial temperature should be high enough such that the acceptance ratio is close to one. The cooling rate α (T_i) in Step 2 of the above algorithm is set to 0.95 [15]. The final temperature is obtained if there is no change in the energy function for five consecutive temperatures. The default number of moves at each temperature was set to 15.

Calculation of Customer Outage Costs in a Distribution Network.

Reliability and hence reliability worth depend on many factors. These factors include failure rates and repair durations of primary and lateral sections, and associated distribution equipment, such as distribution transformers. Other factors are the different types of customers, and their relative location in the network, the load pattern at the load points[28], operating practices such as alternate supply in the event of a contingency, the repair/replacement of distribution transformers. The calculation of outage costs in a distribution system is described in [28] using both analytical and Monte Carlo approaches. A contingency enumeration approach, which considers failures of primary and lateral sections, transformers etc., is used in this paper and is briefly described below.

- 1) Input the outage data, such as component failure and repair rates, and the customer interruption cost data. Consider the system operating practices, such as alternate supply in the event of failures, replacement or repair of failed components, etc.

- 2) Select a load step L_k from the given load model.
- 3) Obtain the indices of failure rate, λ_j , and restoration time (i.e, repair or replacement), r_j , for each outage event j .
- 4) Evaluate the customer cost of interruption, $C_i(r_j)$ (\$/kW) using the corresponding load point customer damage function and outage duration, r_j .
- 5) Evaluate the corresponding expected outage cost (ECOST) due to outage event j using the following equation.
$$\text{ECOST}_{i,j,k} = L_{ki} \cdot C_i(r_j) \cdot \lambda_j \text{ ($/yr.)}$$
, where L_{ki} is the load of step k at load point i , λ_j is the failure rate in occurrence/yr, and r_j is the restoration time in hours.
- 6) Repeat steps 4 and 5 for the number of outage events, n_e , i.e, considering all contingencies on the primary and lateral sections and on distribution transformers which contribute to load point i . The total load point ECOST is then evaluated using the following equation.

$$\text{ECOST}_{i,k} = \sum_{j=1}^{n_e} \text{ECOST}_{i,j,k} \text{ ($/yr)}$$

- 7) Repeat steps 3 to 6 for the number of load points, n_p , in the network. The total ECOST for the network, at the load step L_k is obtained as follows.

$$\text{ECOST}_k = \sum_{j=1}^{n_e} \sum_{i=1}^{n_p} \text{ECOST}_{i,j,k} \text{ ($/yr)}$$

- 8) Repeat steps 2 to 6 for different load steps and obtain total outage cost.

$$\text{ECOST} = \sum_{k=1}^{n_k} \left\{ P_k \sum_{j=1}^{n_e} \sum_{i=1}^{n_p} L_{ik} \cdot C_i(r_j) \cdot \lambda_j \right\}$$

IV. Application

An 11 kV distribution network supplied through a 33/11 kV substation is used to illustrate the algorithm developed to find the optimum routings of feeders and station locations. The peak load of the network is 20 MW with 22 load points supplying energy to 1908 customers. This distribution test system, the failure rates and repair times of the various distribution components together with customer details are provided in [33]. The possible routes of primary and lateral sections and station locations are shown in Figure 2. The corresponding lengths of feeder sections are also shown in Figure 2. It is assumed that all the designated routings in Figure 2 are physically possible. The investment and maintenance costs used in this paper are given in Table 1. The interruption cost data for residential, small industrial and commercial customers are taken from [27] and the data for government and office buildings from [26]. A reasonably simple system is considered in this paper.

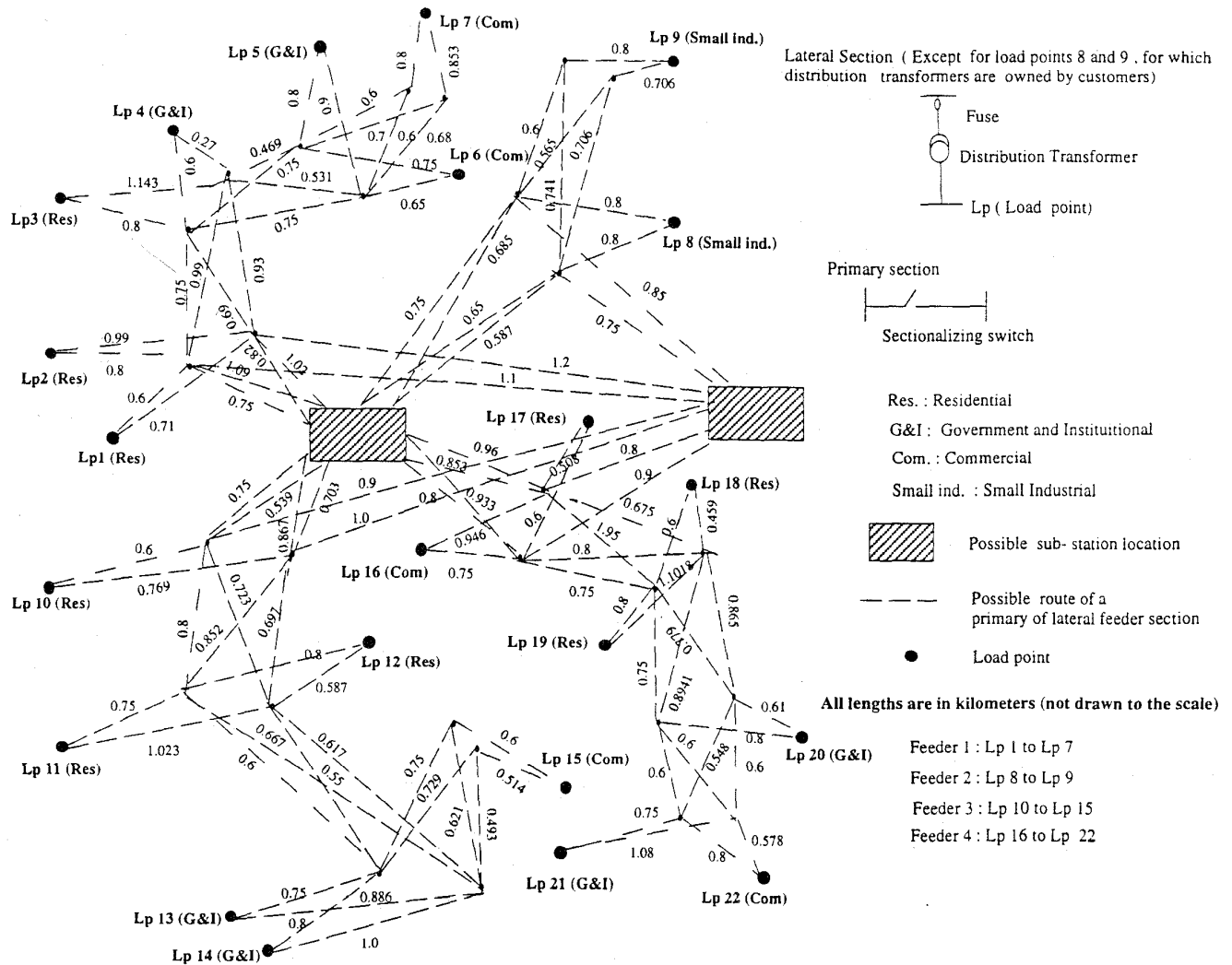


Figure 2. Possible Routes of Distribution Feeder Primary and Lateral Sections and Station Location in the Test System

particularly suited for a large system with many possible feeder routings and station locations.

Table 1 : Investment and Maintenance Cost Related Data

Distribution Equipment	Capital cost	Maintenance cost
11 kV Circuit Breaker	50,000	\$1000/year/Trans
3- ϕ Transformer 500 kVA	\$18064	\$361/year
750 kVA	\$20242	\$405/year
3- ϕ Gang Operated Switch Pad Mounted	\$20337	\$407
Pole Tap, Gang Operated	\$4700	\$94
3- ϕ Fused Cutout	\$610	
3- ϕ Feeder (Urban Over Head)	25000/km	
Life of System Facilities	= 20 years	
Interest Rate	= 8%	

A seven step load duration curve is considered at each load point. The load increment of each step is 10% and probability of the steps are 0.013164, 0.111378, 0.165064,

0.232830, 0.214744, 0.226305, 0.036516. The Feeder i^2r loss coefficient K_c , is set to \$168/kW/yr [32]. The described optimization technique was applied to the system shown in Figure 2. The annual total minimum cost for the system and for the individual feeders is shown in Table 2. The resulting lengths of the primary and lateral sections are also shown in Table 2. The optimal routings of the primary and lateral feeders are shown in Figure 4. This can be compared with Figure 2 which has all feasible routings.

It should be recognized that routing obtained by minimizing a total cost which does not include the outage cost may not lead to the minimum total system cost. In order to illustrate this, the network routing optimization is performed for the given network using simulated annealing without considering the outage costs in the objective function. The total cost is then obtained by computing the outage costs for the resulting network configuration and adding them to the cost obtained by minimizing feeder i^2r loss, investment and maintenance costs. The resulting network configuration which minimizes the investment,

maintenance cost and feeder i^2r costs is shown in Figure 4. The computed total cost for this network and the feeder section lengths are shown in Table 3. The total system minimum cost obtained by including all the costs, i.e., outage, feeder i^2r loss, investment and maintenance costs is \$614640 as shown in Table 2. If the routing is obtained without considering outage costs, the total system cost is \$641801, as shown in Table 3. The optimum routings in each case are different as seen by a careful examination of Figures 3 and 4. The differences can be more clearly appreciated by considering the primary and lateral section feeder lengths shown in Tables 2 and 3.

Table 2. Total Minimum Cost of the Distribution Feeders and the Optimum Routings.

Total System Cost (\$/yr) = 614640		
Feeder	Total Cost obtained by minimizing outage, i^2r loss, invest. and maint. costs (\$/yr)	Lengths of Primary and lateral sections (km)
1	208738	0.75,0.6,0.8,0.75,0.8,0.6,0.75 0.9,0.65,0.6,0.8
2	49646	0.685,0.8,0.565,0.706
3	163863	0.539,0.6,0.723,1.023,0.587,0. 617,0.886,1.0,0.493,0.514
4	194393	0.8,0.75,0.6,0.75,0.6,0.8,0.759, 0.8,0.6,1.08,0.578

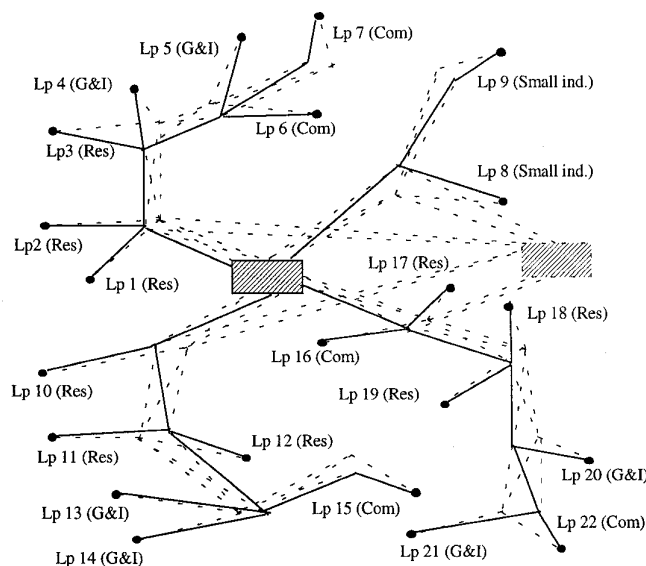


Figure 3. Optimum Routings of the Distribution Feeders Obtained by Minimizing Outage, Resistive loss, Maintenance and Investment Costs.

Customer Composition Sensitivity

The outage costs are influenced by the component failure rates, repair times, customer compositions, operating

practices, and also the relative location of the customers. In order to illustrate the impact of customer location, the customer compositions at load points 3 and 4 have been interchanged. Similarly 5 and 6 have been interchanged.

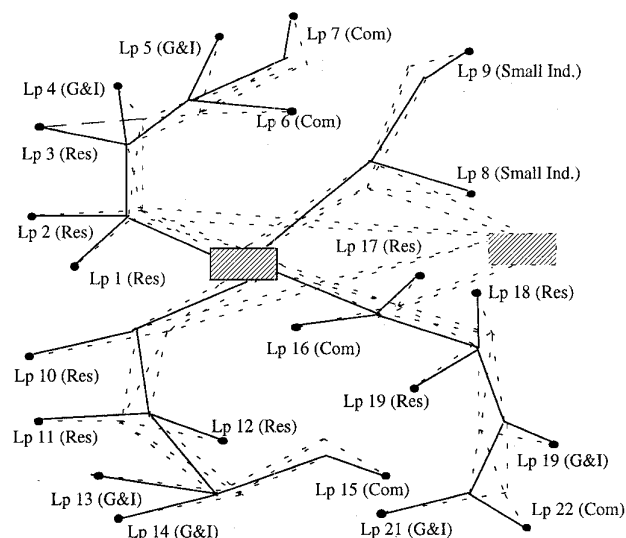


Figure 4. Optimum Routings of the Distribution Feeders obtained by Minimizing Feeder Resistive Loss, Maintenance and Investment Costs.

Table 3. Total cost of the Distribution Feeders Without Including Outage Costs in the Optimization Process.

Total System Cost (\$/yr) = 641801			
Feeder	Minimum Cost obtained minimizing i^2r loss, invest. and maint. costs (A) (\$/yr)	Total cost of feeders (A+Outage costs) (\$/yr)	Lengths of Primary and lateral sections (km)
1	98111	211709	0.75,0.6,0.8,0.75,0.8,0.6,0.75,0.8,0.75,0.6,0.8
2	43513	49646	0.685,0.8,0.565,0.706
3	81080	172218	0.539,0.6,0.723,1.023,0.587,0.55,0.75,0.8,0.729,0.514
4	94210	208228	0.8,0.75,0.6,0.75,0.6,0.8,0.779,0.610,0.548,0.75,0.8

The optimum routing of Feeder 1 in this case is shown in Figure 5a. If the optimum routing is obtained neglecting the outage costs, the configuration is as shown in Figure 5b. The differences can clearly be seen in these two figures. It can also be shown that various operating practices such as alternate supply can influence the outage costs and hence the system configuration.

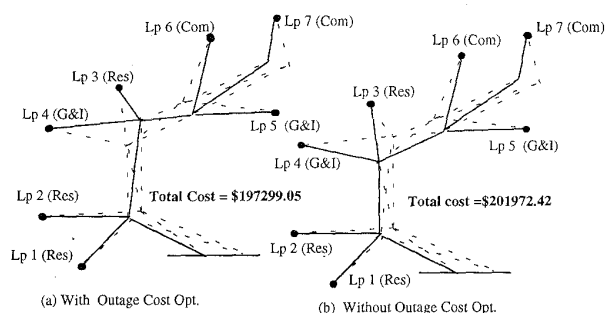


Figure 5. Optimum Routing of Distribution Feeder 1 With Selected Load Point Relocation With and Without Considering Outage Costs in the Optimization Procedure

VII. CONCLUSIONS

A new approach to the optimum distribution feeder routing problem is presented in this paper. This approach not only minimizes resistive, maintenance and investment costs but also includes outage costs in order to obtain the total minimum cost. The paper illustrates that if outage costs are not included in the overall global optimization, the resulting routing could result in higher outage costs and hence higher total system costs. A general purpose combinatorial optimization technique based on simulated annealing approach is used to solve this nonlinear optimization problem.

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